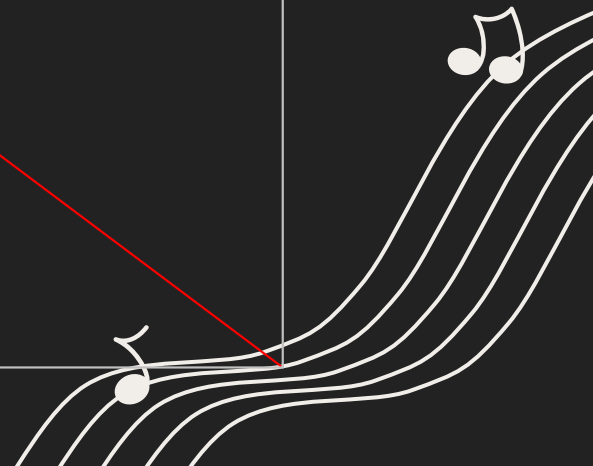
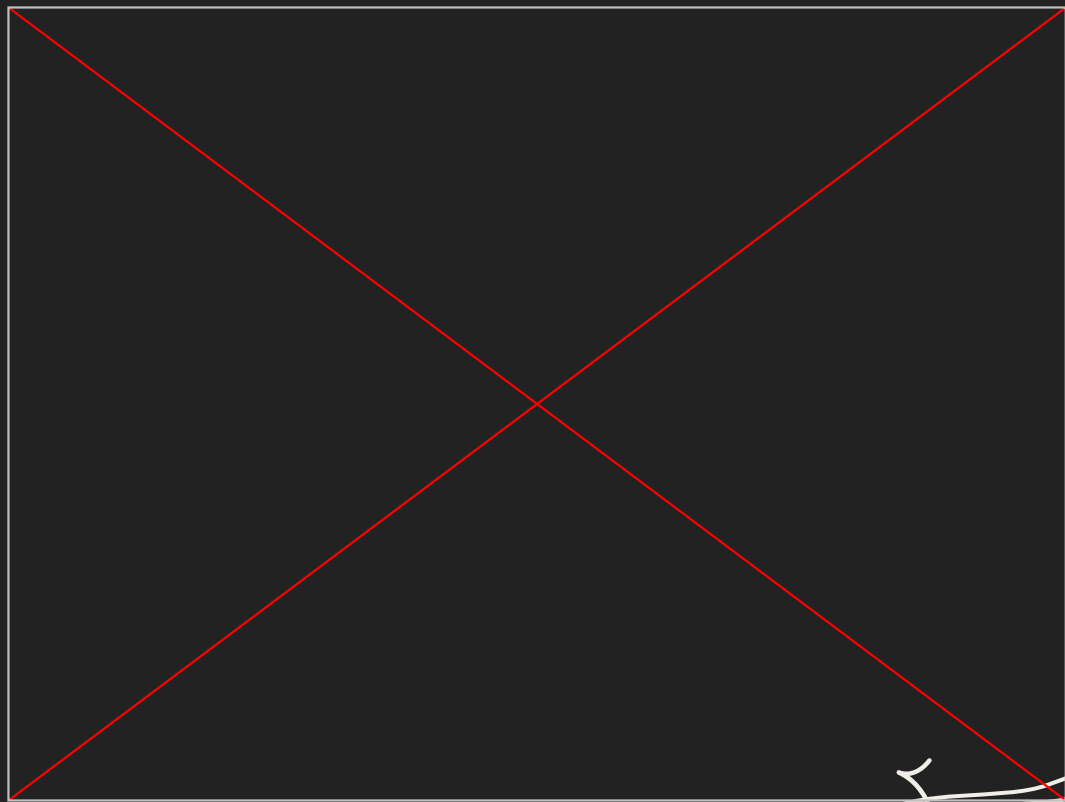
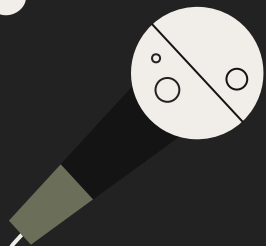


Palate Cleanser

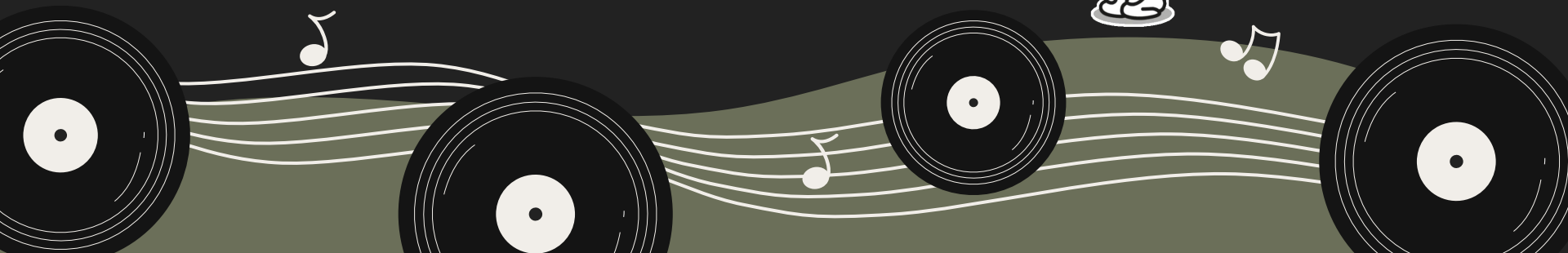




CS35 Final Presentation



Eunice Lee & Jiselle Nguyen



Motivation

Music-inspired dataset from Eunice's Statistics class

Top Spotify songs from 2010-2019 - BY YEAR

▲ 399

<> Code

Download

Data Card

Code (50)

Discussion (6)

Suggestions (0)

Detail

Compact

Column

10 of 15 columns

About this file

There are the most popular songs in the world by year and 13 variables to be explored

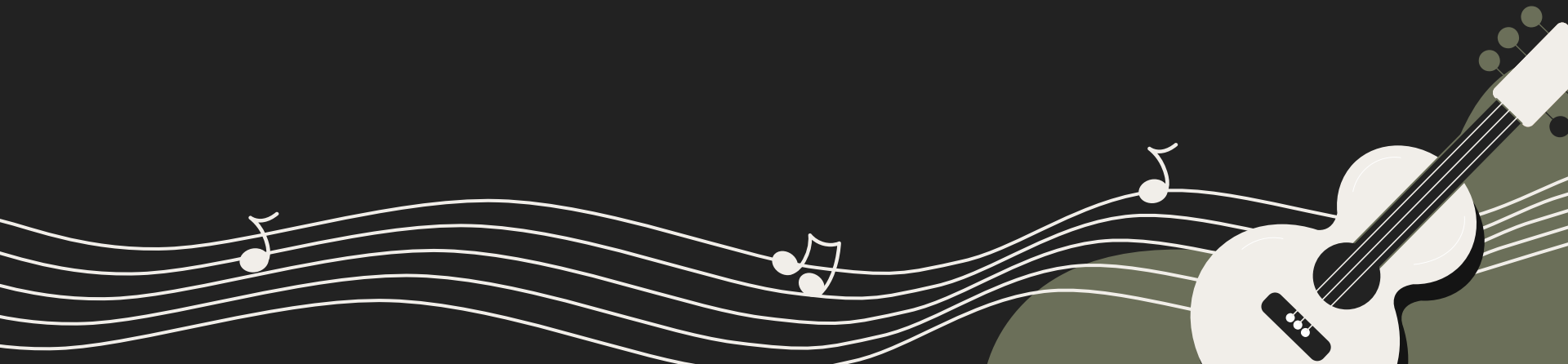
#	title	artist	top genre	year
ID	Song's title	Song's artist	the genre of the track	Song's year i Billboard
1 603	584 unique values	Katy Perry3% Justin Bieber3% Other (570)95%	dance pop54% pop10% Other (216)36%	2010
1	Hey, Soul Sister	Train	neo mellow	2010
2	Love The Way You Lie	Eminem	detroit hip hop	2010
3	TiK ToK	Kesha	dance pop	2010
4	Bad Romance	Lady Gaga	dance pop	2010

Summary

- 1 file
- 15 columns

Big-Picture Problem

- Original idea: make a “Digital Composer”
- Just like how AI can predict the next word in a sentence, wanted to train a model to “hear” a short musical phrase and then continue it
 - Ex: Essentially like teaching a computer to learn the “logic” of a genre so it can finish the song for you



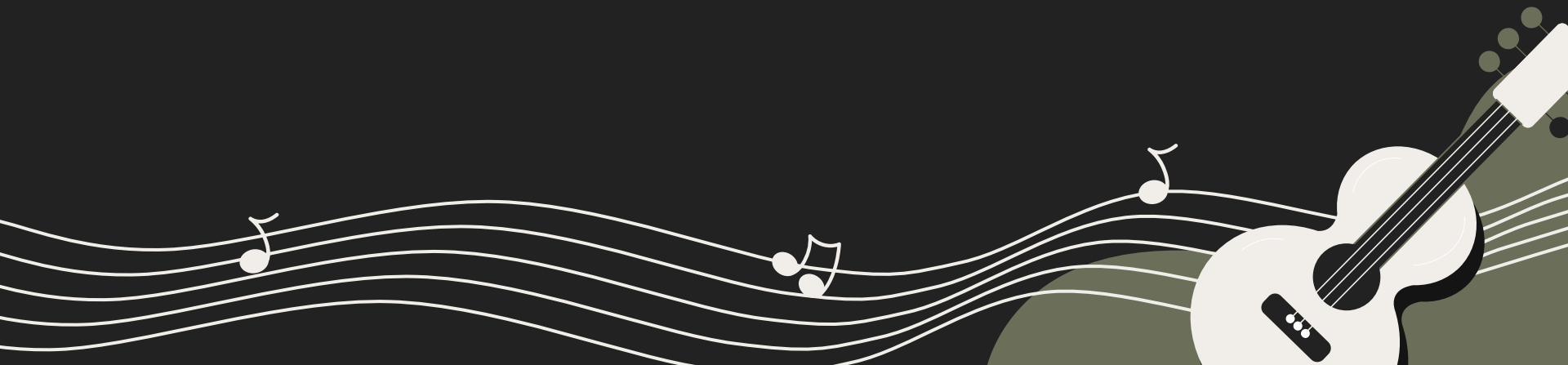
Big-Picture Problem

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- Just like how AI can predict the next word in a sentence, wanted to train a model to “hear” a short musical phrase and then continue it
 - Ex: Essentially like teaching a computer to learn the “logic” of a genre so it can finish the song for you



Big-Picture Problem

- **Our Problem:**
 - Our original model -> outputs being repetitive, boring melodies without variety.
- **Main Goal:** Want to make it under OUR control!!

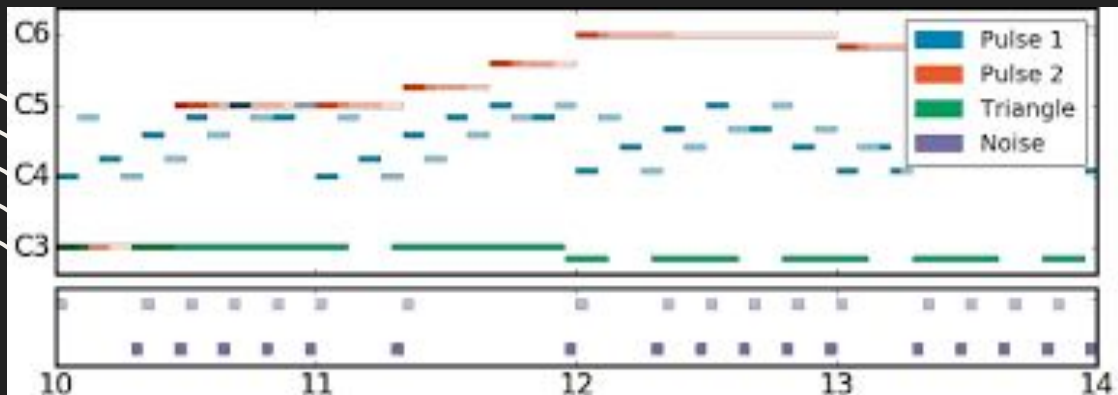


Data!

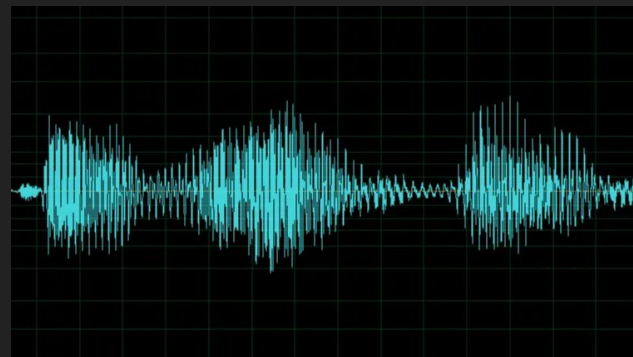
Nintendo®

NES - MDB

~5,000 MIDI files from 397 games



vs.



Libraries!

pretty_midi

scikit-learn

numpy-pandas

Read / write MIDI files

MLPClassifier

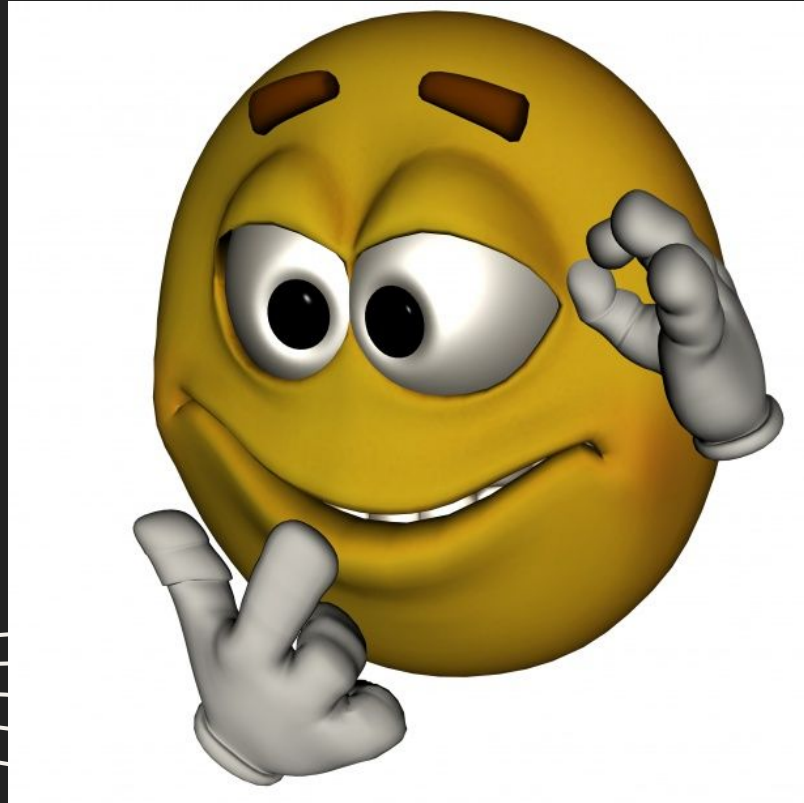
training

- MIDI > Python
- 4 characterizations

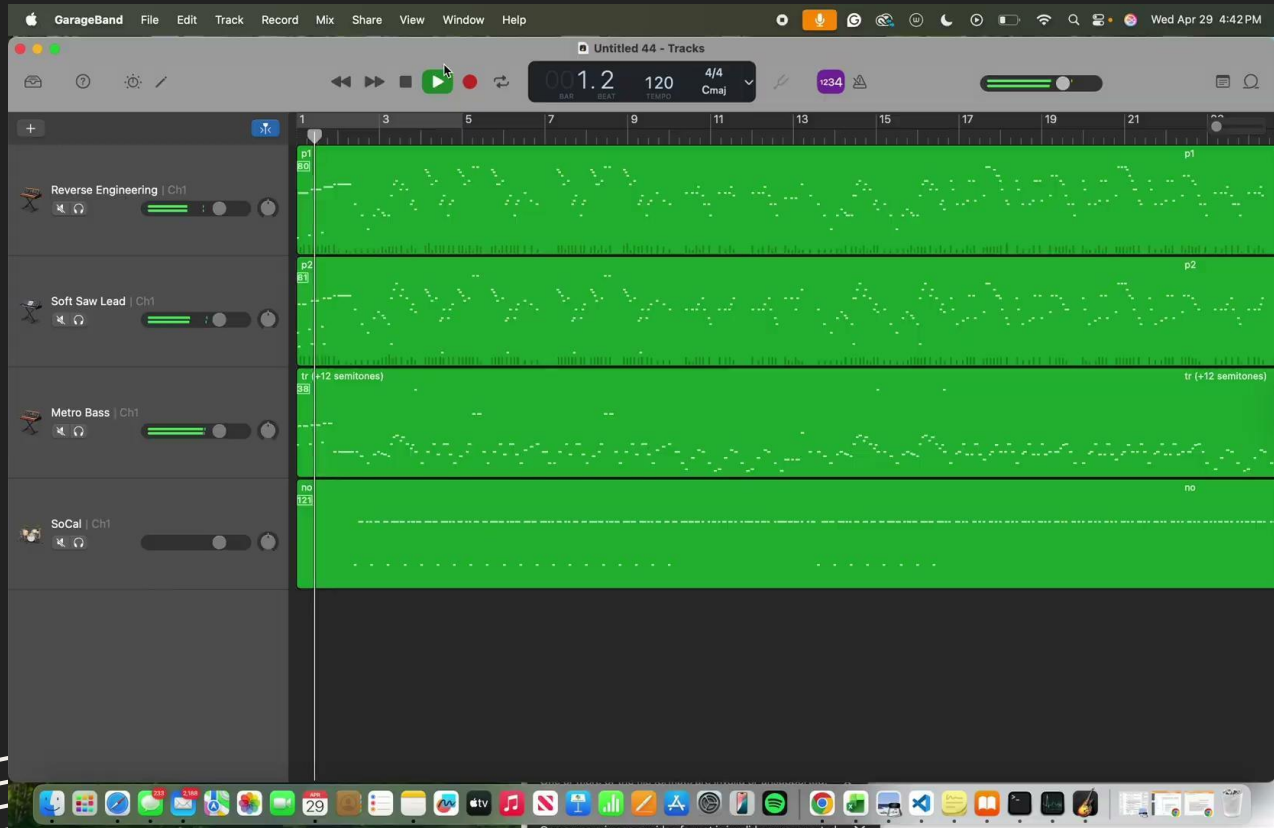
- Categorical
- Next-note predictor

- Flatten CSV
- 858,923- row training CSV
- Temperature, top-k filtering, repetition penalty

Algorithm?



SuperMarioBros “RunningAboutHurry”



Demo

```
from snippet_generator import SnippetGenerator
gen = SnippetGenerator(model_path="output/model.pkl")

gen.generate_snippet(
    key="A", mode="minor", tempo=140,
    variability=0.7, voices=2,
    dynamics="crescendo", note_density=1.5,
    output_path="output/demo.mid"
)
```



Real World Implication

<https://miditoolbox.com/score>



Conclusion



Future plans:

- **Interactive Dashboard:** build a full Graphical User Interface (GUI). Instead of using a terminal, users will have sliders to visually adjust the music in real-time.

Reflection:

- **If we could re-run the past few weeks, we would have pivoted to the probability-based sampling sooner. The "Bass Trap" taught us that raw data often has biases...**

